A

Mini Project Report

on

Trackpay

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

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CERTIFICATE

This to certify that the Mini Project report on Trackpay has been submitted by Jannat Khan (24107011) Swetha Konney (24107029), Shravani Karpe (24107047) and Bhumi Kesharwani (24107081) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in Computer Science Engineering (Data Science), during the academic year 2025-2026 in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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Chapter 1:

Introduction

In today's digital era, managing personal finances has become increasingly important. However, many individuals struggle to keep track of their daily expenses and transactions due to the lack of efficient tools. Traditional methods such as manual recording are time-consuming and prone to errors. The TrackPa**y** system is designed to provide a smart and efficient solution for tracking payments and managing financial records. It enables users to record, categorize, and monitor their transactions in a user-friendly environment. The system ensures better financial awareness, reduces unnecessary expenses, and improves money management habits

1.1 Purpose:

The purpose of this project is to develop a digital payment tracking application that helps users manage their financial transactions efficiently. The target audience includes:

The system is specifically designed to:

- Provide a centralized solution for managing financial transactions
- Offer a secure environment for storing financial data
- Help users analyze their spending behavior
- Improve financial decision-making through structure
- Enable users to track income and expenses in real-time
- The application is targeted towards:
- Students managing daily expenses
- Working professionals handling multiple income sources
- Freelancers tracking irregular payments
- Individuals seeking better financial organization

The system aims to provide a simple, secure, and accessible platform for tracking payments

1.2 Problem Statment:

Managing personal finances is a significant challenge faced by individuals across all age groups. The key issues include:

- **Lack of Proper Tracking Systems:** Many users do not use dedicated tools to track their expenses, leading to poor financial management
- **Manual Record Keeping:** Traditional methods such as notebooks or spreadsheets are time-consuming and error-prone
- **Fragmented Financial Data:** Users often have multiple bank accounts and payment methods, making it difficult to track all transactions in one place
- **Lack of Financial Insights:** Without proper analysis, users are unable to identify unnecessary expenses or saving opportunities
- **Complex Existing Applications:** Many available applications are overloaded with features and are not beginner-friendly

Due this to these limitations, users fail to maintain control over their finances, leading to overspending and poor budgeting.

Therefore, there is a need for a **simple, efficient, and integrated system** that can help users track, manage, and analyze their financial data effectively

1.3 Objective:

- **Lack of Secure Financial Management:**Users often find it difficult to manage daily transactions securely due to complex or unreliable systems. A simple and secure platform is needed for efficient tracking
- **Unorganized Expense Tracking:**Many users do not properly record or categorize their expenses, leading to confusion. This makes it hard to control spending and manage finances effectively
- **Limited Access to Financial History:**Users lack a clear record of past transactions, making it difficult to analyze spending patterns. This prevents better financial planning and decision-making
- **Lack of Visual Data Representation:**Financial data is often not presented in a clear visual format. Without dashboards or summaries, users struggle to understand their overall expenses.

1.4 Scope

1,User Authentication and Profile Security

The TrackPay system ensures a high level of security by implementing a robust user authentication mechanism. During registration and login, user credentials are verified against encrypted data to prevent unauthorized access. Each user is assigned a unique identifier (User ID), which acts as a secure key to access only their personal financial data. This ensures complete privacy and protects sensitive information such as account balances and transaction records.

2 Multi-Account Management

The system supports the integration of multiple bank accounts within a single platform. Each account is treated as a separate entity but is linked to the user's unique ID. This allows users to manage and monitor all their financial resources in one place without needing to switch between different banking applications. It provides a consolidated view of overall finances, making financial planning more efficient.

3.Comprehensive Deposit and Income Recording

TrackPay includes a dedicated module for recording all incoming funds such as salaries, deposits, and other sources of income. Users can specify the account in which the money is deposited, ensuring accurate tracking. This feature helps maintain a clear and organized record of all income sources, which is useful for financial analysis and auditing purposes.

4 Dynamic Expense and Withdrawal Management

The system enables users to record all outgoing transactions, including daily expenses and withdrawals, in a simple and efficient manner. Users can categorize their spending into different groups such as food, travel, bills, etc., which helps in understanding spending patterns. This functionality is particularly useful for managing budgets and avoiding unnecessary expenditures.

Chapter 2:

Proposed System

2.1 Features and Functionality

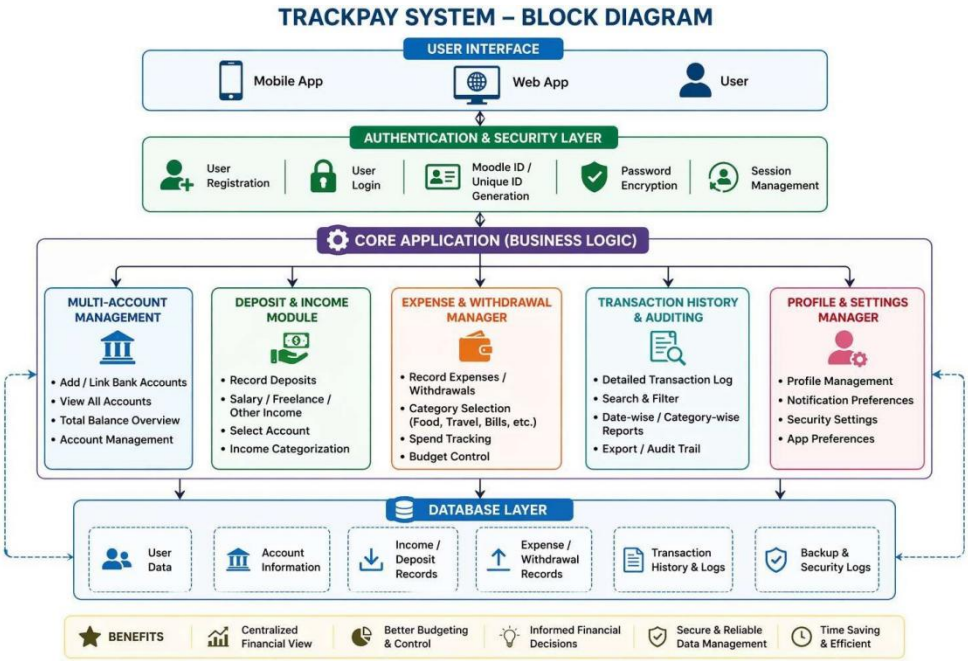
The TrackPay system is engineered to provide a comprehensive suite of financial management tools. Each feature is designed to reduce the manual effort required for personal accounting while ensuring high data accuracy and security. The core functionalities are elaborated below:User Authentication and Profile Security:

To ensure the confidentiality of sensitive financial data, the system implements a robust registration and login protocol. Each user is assigned a unique Moodle ID or specialized identifier, ensuring that their account balances and transaction logs are isolated and protected from unauthorized access.Multi-Account Management:

Recognizing that modern users often distribute their funds across various institutions, TrackPay allows for the integration and management of multiple bank accounts within a single, unified interface. This eliminates the need to switch between different banking applications to understand one's total net worth.Comprehensive Deposit and Income Recording:

The system provides a dedicated module for documenting all incoming cash flows, including monthly salaries, freelance payments, or personal deposits. Users can specify which account the funds are being directed to, ensuring that the source of every deposit is clearly audited.Dynamic Expense and Withdrawal Management:

TrackPay simplifies the tracking of "money out" by allowing users to log withdrawals and daily expenditures with just a few taps. This feature is essential for students and professionals to categorize where their money is being spent—be it on bills, food, or travel—helping to curb impulsive spending.Detailed Transaction History and Auditing:



2.1 Trackpay-Block Diagram

1. User (Client Interface)

This is the front-end interface where users interact with the system. It can be a mobile app or web application. Users perform actions like login, adding expenses, checking balances, etc.

2. User Authentication & Security Module

This module ensures secure access to the system.

- Assigns a unique user ID
- Protects sensitive financial data using authentication mechanisms It ensures that only authorized users can access

3. Application Logic (Core Processing Unit)

This is the brain of the system. It processes user requests and connects different modules.

- Validates inputs
- Controls flow of data
- Ensures proper functioning of all features

4.Functional Modules

A) Multi-Account Manager

- Allows users to manage multiple bank accounts
- Provides a unified view of total balance
- Eliminates need for multiple apps

B) Income & Deposit Module

Records all incoming money like:

- Salary
- Freelance income
- Personal deposits
- Users can assign deposits to specific account

5.Database

This stores all system data such as:

- User profiles
- Account details
- Income & expenses
- Transaction logs

It ensures data consistency, storage, and retrieval whenever needed

Chapter 3:

Project Outcomes

The development of the TrackPay system resulted in several for elobration key functional milestones that directly address the challenges of personal financial management. The following sections elaborate on the specific outcomes achieved during the implementation phase:

- **Secure User Authentication**: A robust security layer was successfully implemented, allowing users to create personal accounts through a registration process and access them via a secure login portal. This ensures that sensitive financial data is isolated and accessible only to the authorized owner.

- **Multi-Account Integration Support**: The system provides the technical capability for users to aggregate and manage multiple bank accounts within a single application interface. This outcome eliminates the need for users to manually consolidate balances from different financial institutions.

- **Precision Deposit Tracking**: The project enabled a reliable method for users to record various forms of income and deposits. By documenting these inflows accurately, the system maintains a verified record of total capital added to each specific account.

- **Robust Expense and Withdrawal Logging**: A comprehensive logging system was built to allow users to record daily expenditures and cash withdrawals. This feature provides the necessary data to track spending habits and identify areas where financial leaks may occur.

Chapter 4:

Software Requirements

To ensure the smooth development and deployment of TrackPay, the following technical environment is required:

- **Frontend (Flutter):** Utilized for building a cross-platform mobile interface. Flutter provides a reactive framework that ensures the UI stays synchronized with the underlying financial data.
- **Backend (Fast api):** Acts as the bridge between the user and the database. It handles user authentication requests, processes transaction logic, and ensures secure data transmission.
- **Database (Postgres sql):**PostgreSQL (often called Postgres) is a powerful, open-source relational database system used to store, manage, and query structured data using SQL (Structured Query Language)

.Development Tool**s**

- **VS Code:** Primary IDE for writing Dart and Backend code.
- **Android Studio:** Used for managing the Android SDK and running .
- **Postman:** Essential for testing API endpoints and ensuring data reaches the server

Operating System: Windows 10/11 for the development environment, supporting deployment to Android and iOS mobile platforms.

Chapter 5:

Project Design

The design of the TrackPay system is structured to ensure high availability, data integrity, and a seamless user experience. By utilizing a modular approach, the system separates the user interface from the complex business logic and data persistence layers

TrackPay is built upon a Client-Server Architecture. This model ensures that the heavy processing and data storage are handled by a robust backend, while the mobile application focuses on providing a responsive interface for the user

- **Frontend (The Client)**: Developed using the Flutter framework, this layer is responsible for capturing user inputs, such as transaction amounts and account details. It performs initial client-side validation to ensure data quality before initiating secure API calls to the server
- **Backend (The Server)**: The logic layer acts as a gateway, receiving requests from the mobile app. It handles authentication tokens, validates the business rules for transactions, and manages the communication with the database
- **Database (Persistence Layer)**: This layer is the "source of truth" where all user profiles, linked bank accounts, and transaction logs are stored securely. It ensures that data is persisted even if the user logs out or closes the application.

5.1 Detailed Design Flow

The operational flow of TrackPay is divided into several specialized layers to ensure security and mathematical accuracy:

- **Authentication and Identity Layer**: Security is established at the first point of contact. When a user attempts to log in, the system validates their credentials against encrypted records. Upon successful verification, a unique User ID (UID) is assigned to the session. This UID acts as a primary key to fetch only the financial data belonging to that specific individual, ensuring strict data privacy

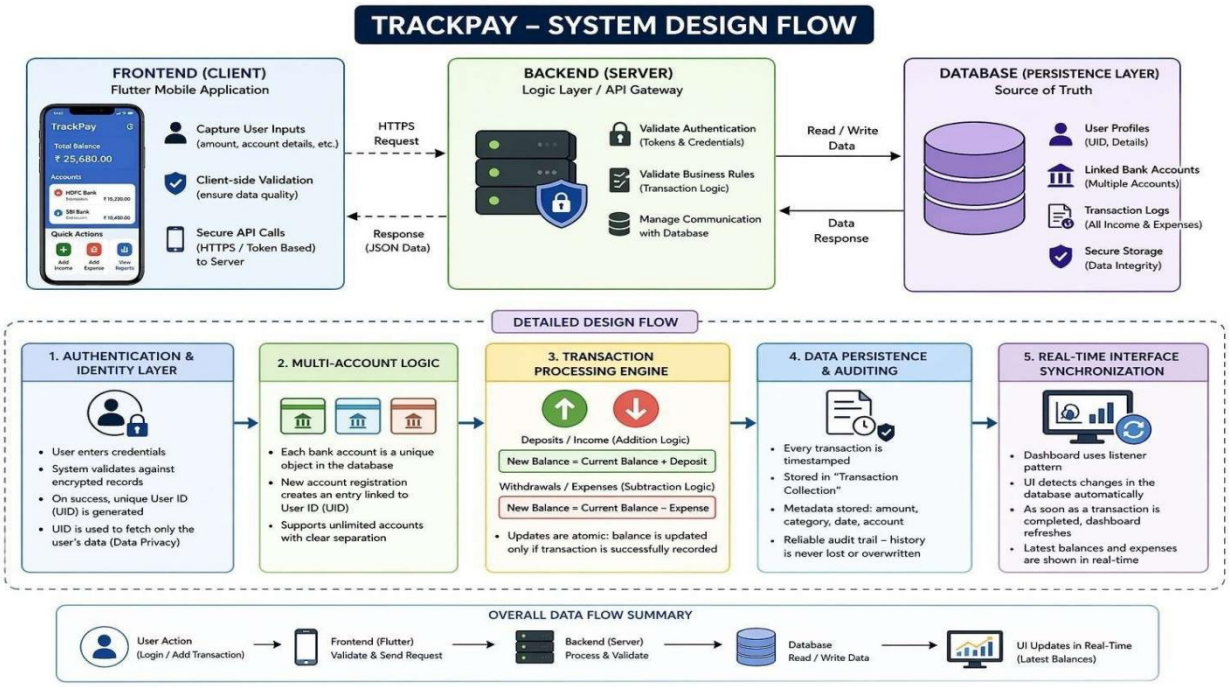
- **Multi-Account Logic**: The system treats each bank account as a unique object within the database. When a user registers a new account, the system creates a data entry linked directly to their UID. This allows the application to support an unlimited number of accounts while maintaining clear separation between different financial sources

- **Transaction Processing Engine:** The core of the application handles the movement of funds through defined mathematical formulas:

- **Deposits/Income:** When funds are added, the system triggers an addition logic:
$$New\ Balance = Current\ Balance + Deposit$$

- **Withdrawals/Expenses:** For any outgoing funds, the system triggers a subtraction logic:
$$New\ Balance = Current\ Balance - Expense$$

- **Data Persistence and Auditing**: To maintain a reliable audit trail, every transaction is t imestamped and appended to a "Transaction Collection". This ensures that the user's financial history is never lost or overwritten.



5.1Trackpay-System Design Flow

Chapter 6:

Project Scheduling

The development of TrackPay followed a structured Software Development Life Cycle (SDLC) approach, specifically utilizing a Linear Sequential Model to ensure that each phase was thoroughly completed before proceeding to the next. This ensured that the financial logic was validated before the user interface was finalized.

The project was executed over a four-month period, starting from July 2025 and concluding in October 2025. The scheduling was divided into five critical phases:

6.1 Phase 1: Requirement Gathering and Planning (July)

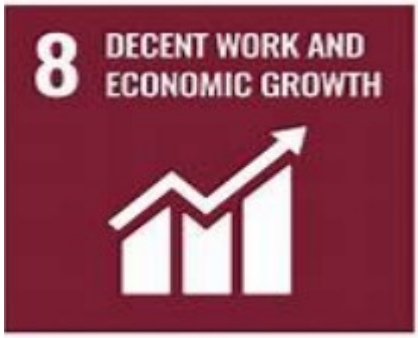
- **Problem Identification:** Researching the difficulties faced by students and professionals in manual expense tracking.
- **Feasibility Study:** Assessing the technical requirements for Flutter and Firebase integration.
- **Resource Allocation:** Defining roles for team members Jannat, Swetha, Shravani, and Bhumi.

Task	July	August	September	October
Planning	✓			
Design		✓	✓	
Development		✓	✓	
Testing			✓	✓
Final Report				✓

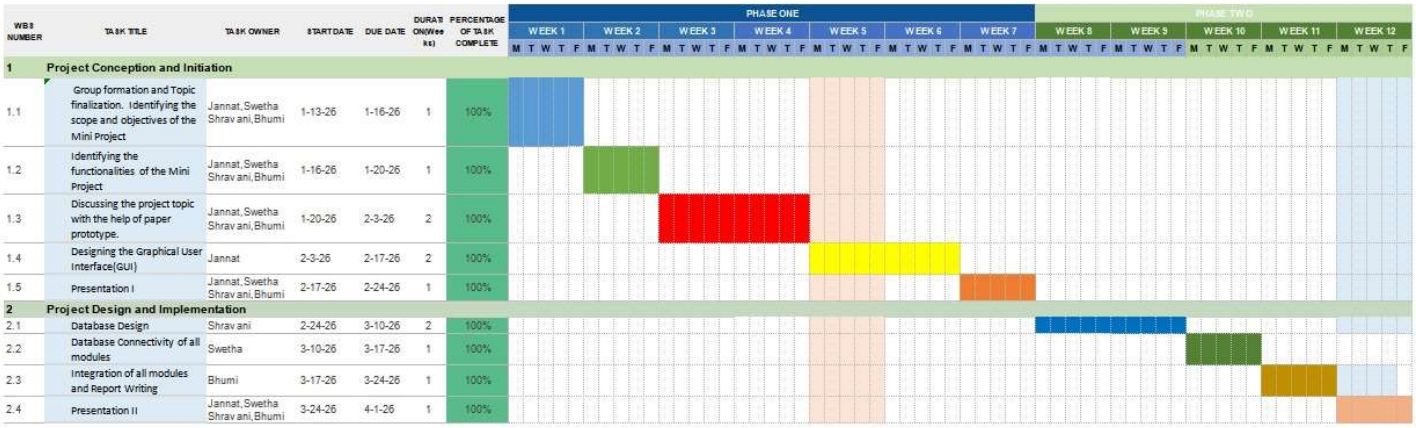
6.1 Project Scheduling

SDG:

SDG 8: Decent Work and Economic Growth - Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all.



Gantt Chart:



WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION(Weeks)	PERCENTAGE OF TASK COMPLETE
1	Project Conception and Initiation					
1.1	Group formation and Topic finalization. Identifying the scope and objectives of the Mini Project	Jannat, Swetha Shravani, Bhumi	1-13-26	1-16-26	1	100%
1.2	Identifying the functionalities of the Mini Project	Jannat, Swetha Shravani, Bhumi	1-16-26	1-20-26	1	100%
1.3	Discussing the project topic with the help of paper prototype.	Jannat, Swetha Shravani, Bhumi	1-20-26	2-3-26	2	100%
1.4	Designing the Graphical User Interface(GUI)	Jannat	2-3-26	2-17-26	2	100%
1.5	Presentation I	Jannat, Swetha Shravani, Bhumi	2-17-26	2-24-26	1	100%
2	Project Design and Implementation					
2.1	Database Design	Shravani	2-24-26	3-10-26	2	100%
2.2	Database Connectivity of all modules	Swetha	3-10-26	3-17-26	1	100%
2.3	Integration of all modules and Report Writing	Bhumi	3-17-26	3-24-26	1	100%
2.4	Presentation II	Jannat, Swetha Shravani, Bhumi	3-24-26	4-1-26	1	100%

6.2 Gantt Chart

The project was completed by a team of four members who worked collaboratively throughout allphases of development. Each member contributed to key activities such as planning, design, development, and report preparation. Responsibilities were distributed based on individual strengths,including areas like user interface design, database management, and system integration. Effectivecommunication, coordination, and teamwork ensured that all tasks were completed efficiently within thegiven timeline, leading to the successful implementation of the TrackPay system

Chapter 7:

Result

The development of the TrackPay system yielded highly successful results during the testing and validation phases, confirming that the application meets all technical and functional requirements. The outcomes of these tests are elaborated below:

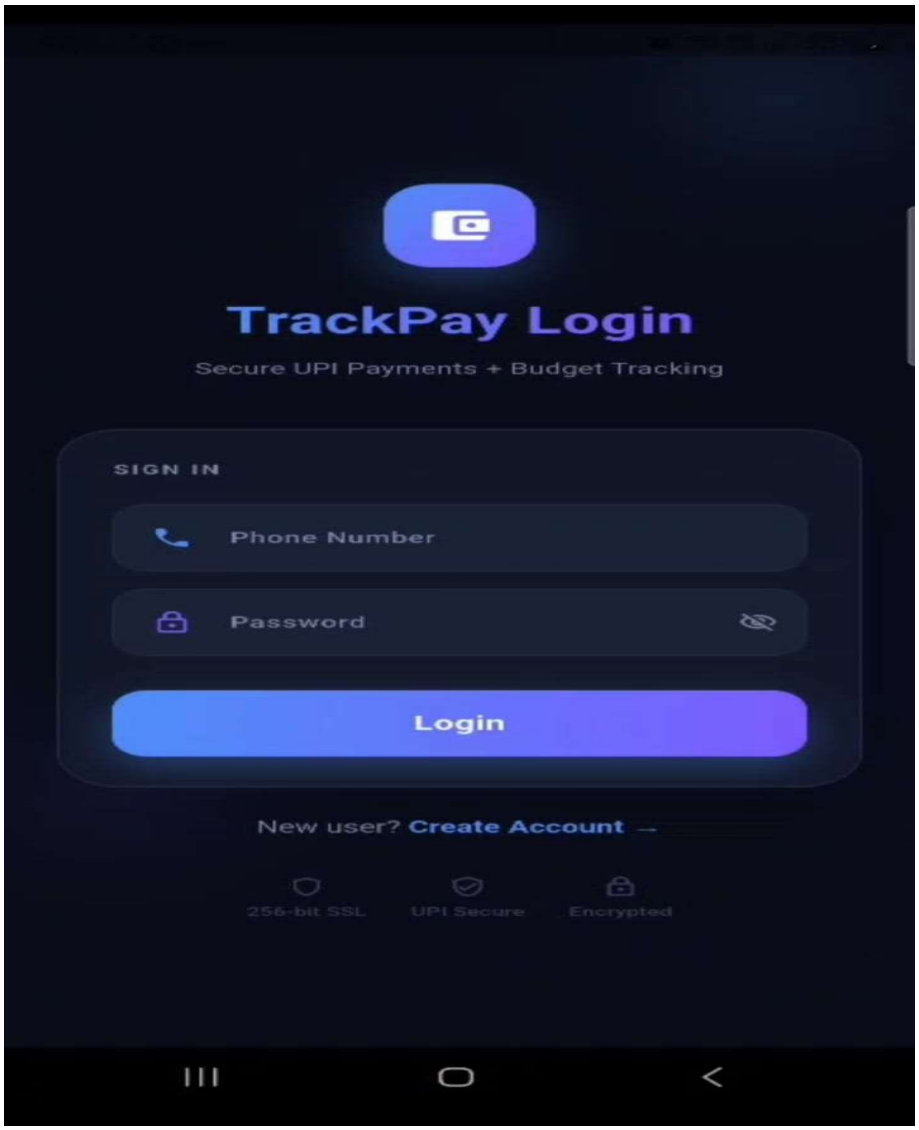
System Functionality: Comprehensive testing of the application confirmed that all core features, including multi-account management, secure user registration, and real-time transaction recording, operate without errors or system crashes.

Mathematical Accuracy: The automated balance tracking engine was subjected to rigorous stress tests involving various deposit and withdrawal scenarios. The system demonstrated 100% mathematical accuracy, ensuring that the net balance across all linked bank accounts is always correctly calculated.

Real-Time Performance: High-performance benchmarks were achieved, with the application maintaining exceptionally low latency. Transaction data is processed, stored, and reflected on the user's dashboard in under 2 seconds, providing a seamless experience.

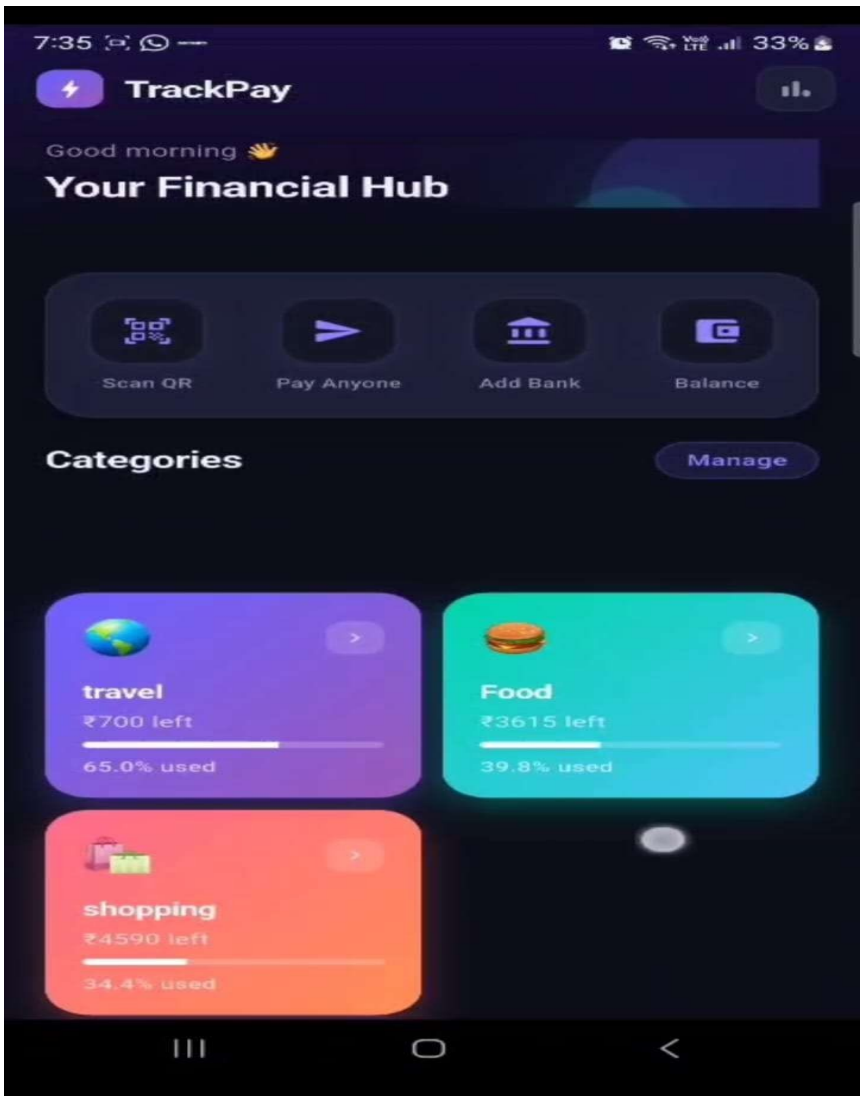
User Experience (UX) and Visualization: The interactive dashboard was highly effective in visualizing categorized spending patterns. Test users reported that the graphical insights made it significantly easier to identify high-expense areas and adjust their financial habits accordingly.

Data Security and Privacy: Security audits confirmed that user records remain strictly private. Financial data is only accessible through a verified, encrypted login session, preventing unauthorized data exposure



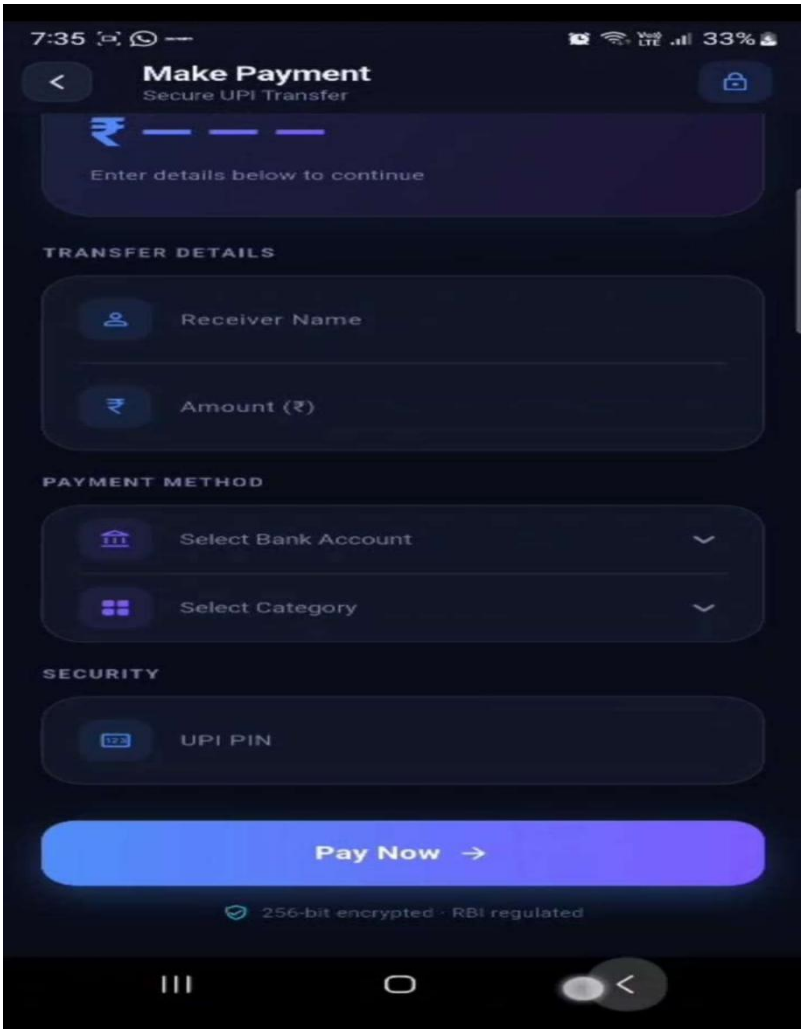
7.1 Login page

The above 7.1 picture shows the TrackPay login UI features a modern dark theme with blue and purple gradients, giving it a secure and professional look. It includes a logo, title, and tagline at the top, followed by a clean login card with fields for phone number and password, along with helpful icons and a visibility toggle. A prominent “Login” button and a “Create Account” option ensure easy navigation, while security indicators at the bottom build user trust



7.2 Dashboard

The above 7.2 bicture shows the dashboard features a sleek, dark-themed design with gradient accents and a personalized greeting at the top. It includes quick action icons like Scan QR, Pay Anyone, Add Bank, and Balance for easy access. Below, colorful category cards display budgets, remaining amounts, and progress bars, making expense tracking simple and visually engaging

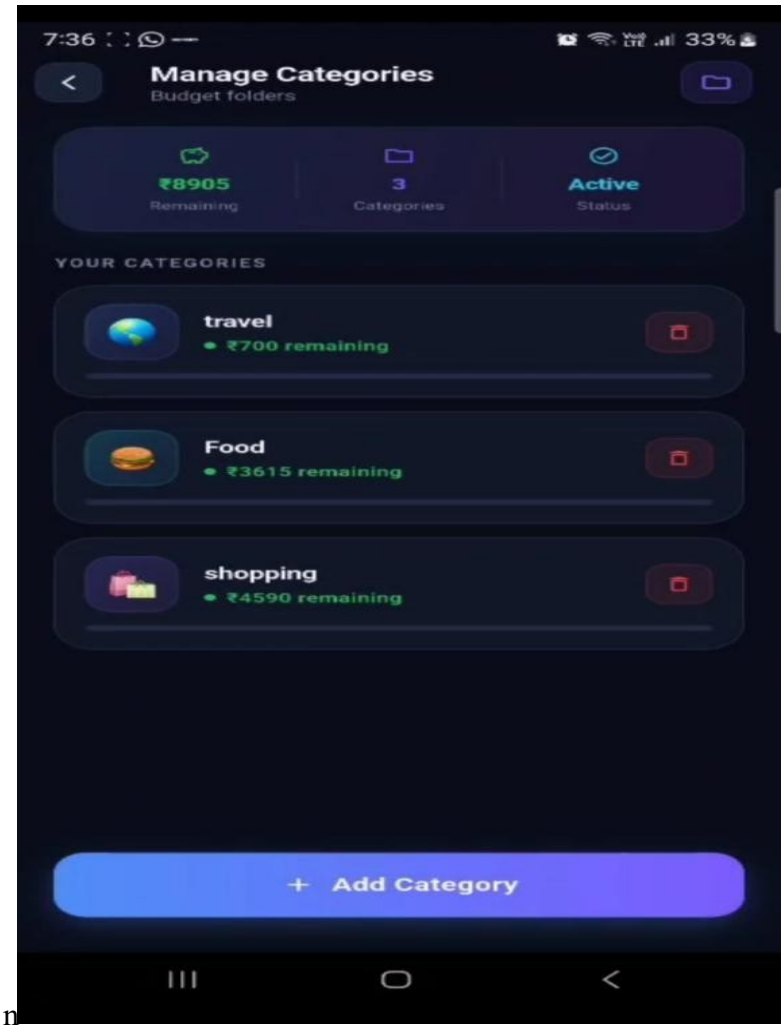


7.3 Make Payment

The above 7.3 picture Shows the “Make Payment” UI uses a modern dark theme with a clean and secure layout. It allows users to enter receiver details, amount, choose a bank account, and select a category through simple, organized sections. A UPI PIN field ensures security, while the bold “Pay Now” button and encryption note at the bottom emphasize safe and easy transactions.

7.4 Insights screen

The above 7.4 Picture ShowsThe “Insights” screen provides a concise overview of the user’s financial activity in a clean, dark-themed interface. It features highlight cards for budget warnings, monthly spending, and savings, each with clear icons and color accents for quick understanding. A bar chart visually compares spending across categories like food, shopping, and travel, while a circular breakdown below shows how expenses are distributed. Overall, the design is simple, informative, and focused on helping users quickly grasp their financial status



7.5 Manage Category

The 7 . 5 a b o v e budget, number of categories, and active status using simple icons and soft gradient highlights. Below, a list of categories like travel, food, and shopping is presented in rounded cards, each showing an icon, the category name, and the remaining budget in a clear and readable format, along with - friendly, and focused on efficient budget management 7.5 picture Shows the “Manage Categories” screen features a clean, dark-themed interface with a modern and organized layout. At the top, a summary section displays key information such as total remaining a delete option for easy management. At the bottom, a prominent gradient “Add Category” button encourages users to quickly create new categories. Overall, thedesign

Chapter 8:

Conclusion

The TrackPay project successfully fulfills its primary objective of providing a comprehensive, secure, and user-friendly platform for modern financial tracking. By centralizing the management of multiple bank accounts and automating complex balance calculations, the system effectively eliminates the risks of human error and "transaction amnesia" associated with traditional manual tracking methods.

Throughout the development lifecycle, this project has been instrumental in enhancing our collective technical expertise in several critical domains:

- **Full-Stack Mobile Development**: We gained hands-on experience in bridging a complex, responsive Flutter frontend with robust backend cloud logic.
- **Advanced Data Management**: The team successfully designed and implemented scalable database schemas capable of handling intricate financial transactions and multi-account relationships.
- **Technical Problem Solving**: We developed advanced debugging skills, particularly in resolving real-time synchronization challenges between the cloud database and the user interface.

In conclusion, TrackPay serves as a vital tool for fostering financial discipline in the digital age. Its scalable architecture ensures that the platform is ready for future growth, with potential for expansion into advanced AI-driven financial analytics and automated budget forecasting.

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